SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Science, Mathematics and Technology

SSILP Math

Beginner

EXPONENTS AND RADICALS

Learning objectives

- Understand expressions such as $a^{m/n}$
- Understand and use negative exponents
- Use rules for integer exponents, radicals, and n^{th} roots for simplifying expressions

Videos

Expressions with exponents (Khan Academy)
Understanding roots (Khan Academy)
Simplifying radicals with variables, exponents, fractions (Organic Chemistry tutor)

Beginner level videos for algebra fundamentals
Skip if you already know this material

Session outline

1. Exponents

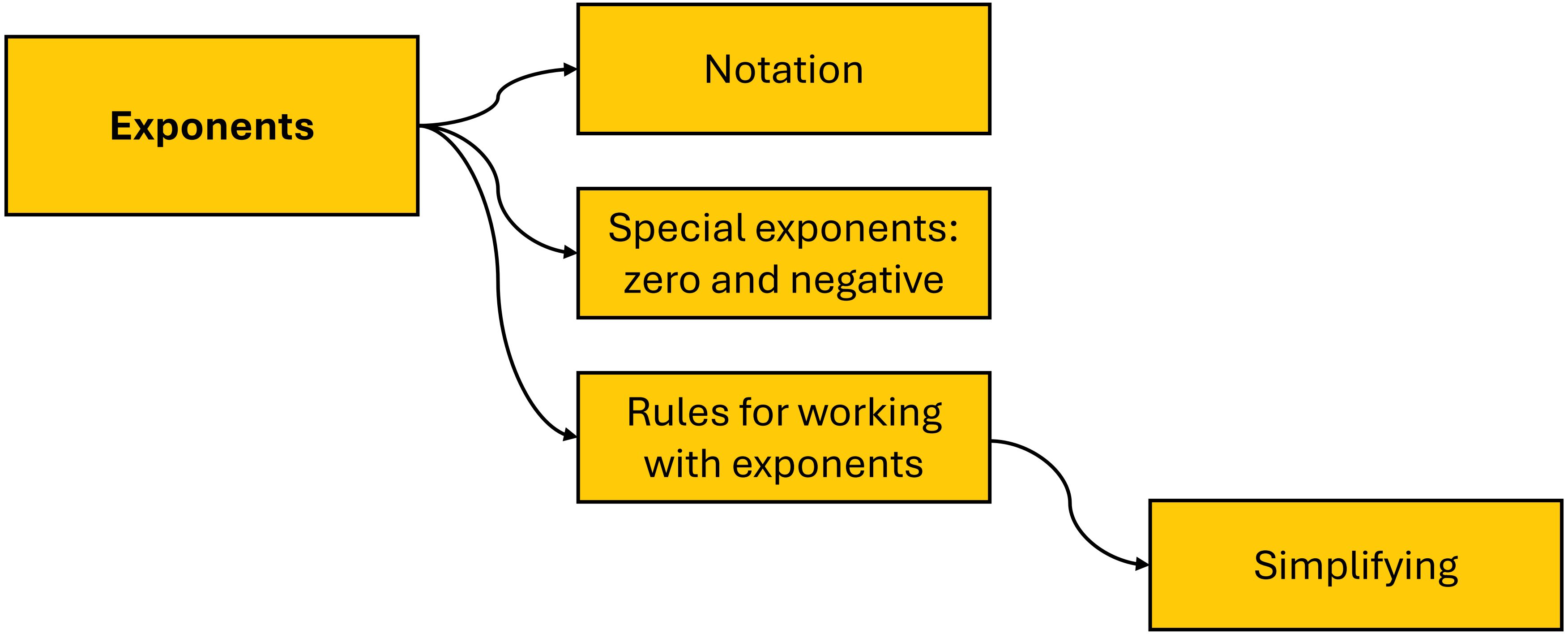
2. Radicals

Chapter 1 of Textbook: FUNDAMENTALS [contd from last session]

1. Exponents

Exponents

Overview



Refer to the following pages in the textbook!

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

Chapter 1, Section 1.2: Exponents and radicals

Pg 14-16

Exponentiation

multiplication is **repeated addition**

$$2 \cdot 3 = 2 + 2 + 2$$

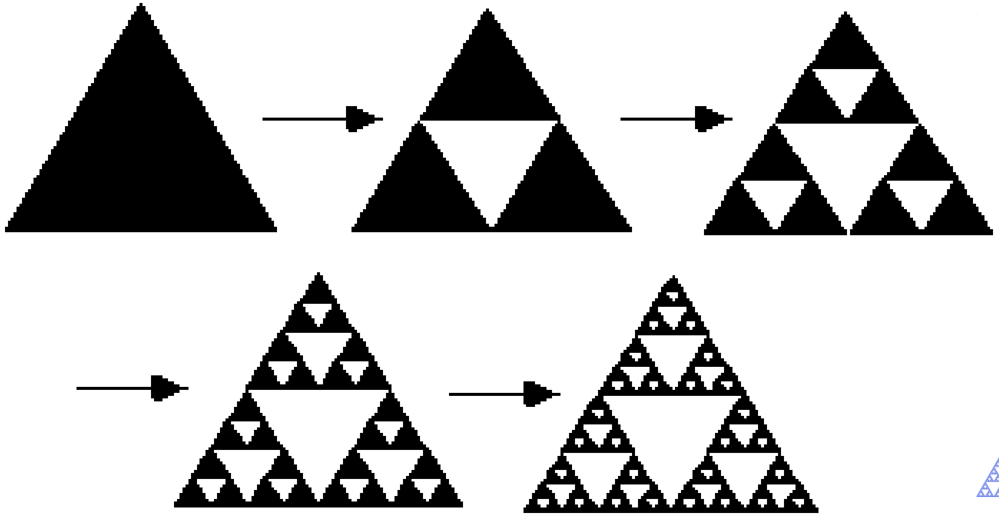
exponentiation is **repeated multiplication**

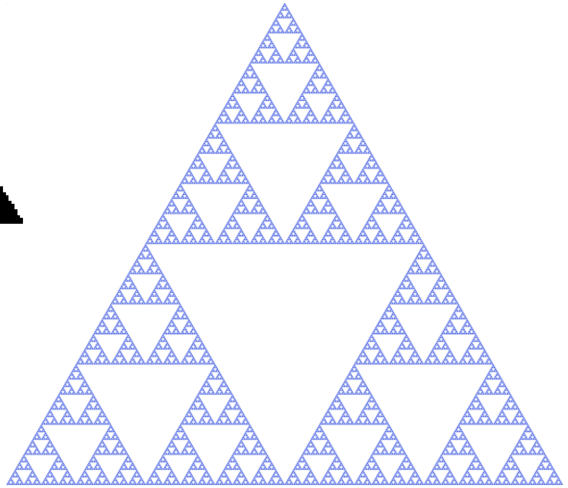
$$2^3 = 2 \cdot 2 \cdot 2$$

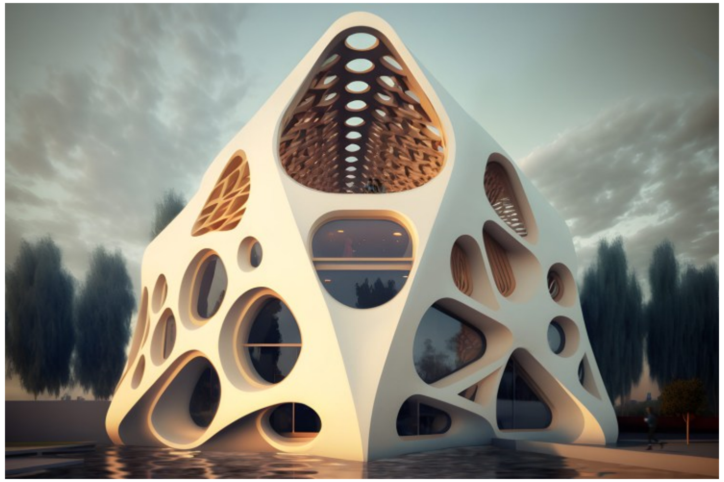
exponents generate **very large numbers**

$$4^8 = 65{,}536$$

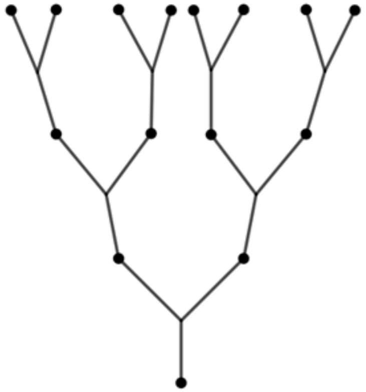
Exponential Growth as inspiration in Design

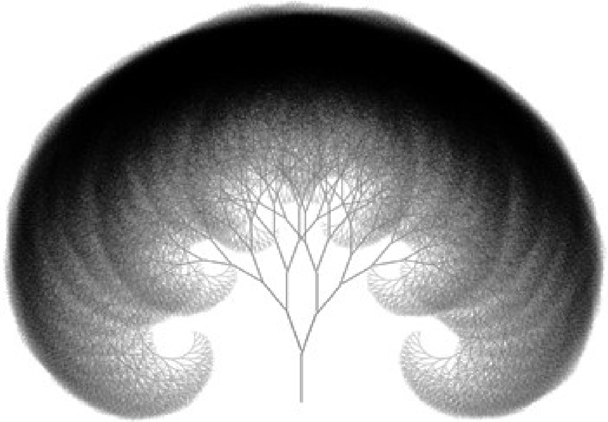






Sierpinski's Triangle





Tree form generated by exponential branching

What is an Exponent?

An exponent or power is a mathematical operation of repeated multiplication.

Base → b^{n} ← Exponent

b^{n} means "b raised to the power of n".

$$5^3 = 5 \times 5 \times 5 = 125$$

Exponent: Formal definition

EXPONENTIAL NOTATION

If a is any real number and n is a positive integer, then the ***n*th power** of a is

$$a^n = \underbrace{a \cdot a \cdot \cdots \cdot a}_{n \text{ factors}}$$

The number a is called the **base**, and n is called the **exponent**.

Review your past knowledge!

Remember how to simplify expressions?

Order of performing operations

PEMDAS

Parentheses (incl. absolute values, brackets, fraction lines, radicals)

Exponents (and roots)

Multiplication & Division

Addition & Subtraction

Mnemonic:

"Please Excuse My Dear Aunt Sally"

Exponential notation

(a) $\left(\frac{1}{2}\right)^5 = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{32}$

(b) $(-3)^4 = (-3)\cdot(-3)\cdot(-3)\cdot(-3) = 81$

(c) $-3^4 = -(3\cdot 3\cdot 3\cdot 3) = -81$

Note the distinction between $(-3)^4$ and -3^4. In $(-3)^4$ the exponent applies to -3, but in -3^4 the exponent applies only to 3.

EXAMPLE

Zero and negative exponents

ZERO AND NEGATIVE EXPONENTS

If $a \neq 0$ is a real number and n is a positive integer, then

$$a^0 = 1 \qquad \text{and} \qquad a^{-n} = \frac{1}{a^n}$$

(a) $\left(\frac{4}{7}\right)^0 = 1$

(b) $x^{-1} = \dfrac{1}{x^1} = \dfrac{1}{x}$

(c) $(-2)^{-3} = \dfrac{1}{(-2)^3} = \dfrac{1}{-8} = -\dfrac{1}{8}$

Laws of exponents

Law	Example	Description
1. $a^m a^n = a^{m+n}$	$3^2 \cdot 3^5 = 3^{2+5} = 3^7$	To multiply two powers of the same number, add the exponents.
2. $\dfrac{a^m}{a^n} = a^{m-n}$	$\dfrac{3^5}{3^2} = 3^{5-2} = 3^3$	To divide two powers of the same number, subtract the exponents.
3. $(a^m)^n = a^{mn}$	$(3^2)^5 = 3^{2 \cdot 5} = 3^{10}$	To raise a power to a new power, multiply the exponents.
4. $(ab)^n = a^n b^n$	$(3 \cdot 4)^2 = 3^2 \cdot 4^2$	To raise a product to a power, raise each factor to the power.
5. $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$	$\left(\dfrac{3}{4}\right)^2 = \dfrac{3^2}{4^2}$	To raise a quotient to a power, raise both numerator and denominator to the power.
6. $\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$	$\left(\dfrac{3}{4}\right)^{-2} = \left(\dfrac{4}{3}\right)^2$	To raise a fraction to a negative power, invert the fraction and change the sign of the exponent.
7. $\dfrac{a^{-n}}{b^{-m}} = \dfrac{b^m}{a^n}$	$\dfrac{3^{-2}}{4^{-5}} = \dfrac{4^5}{3^2}$	To move a number raised to a power from numerator to denominator or from denominator to numerator, change the sign of the exponent.

Simplifying expressions with exponents

Simplify:

(a) $(2a^3b^2)(3ab^4)^3$ **(b)** $\left(\frac{x}{y}\right)^3\left(\frac{y^2x}{z}\right)^4$

SOLUTION

(a)
$$\begin{aligned}(2a^3b^2)(3ab^4)^3 &= (2a^3b^2)[3^3a^3(b^4)^3] && \text{Law 4: } (ab)^n = a^nb^n\\ &= (2a^3b^2)(27a^3b^{12}) && \text{Law 3: } (a^m)^n = a^{mn}\\ &= (2)(27)a^3a^3b^2b^{12} && \text{Group factors with the same base}\\ &= 54a^6b^{14} && \text{Law 1: } a^ma^n = a^{m+n}\end{aligned}$$

(b)
$$\begin{aligned}\left(\frac{x}{y}\right)^3\left(\frac{y^2x}{z}\right)^4 &= \frac{x^3}{y^3}\frac{(y^2)^4x^4}{z^4} && \text{Laws 5 and 4}\\ &= \frac{x^3}{y^3}\frac{y^8x^4}{z^4} && \text{Law 3}\\ &= (x^3x^4)\left(\frac{y^8}{y^3}\right)\frac{1}{z^4} && \text{Group factors with the same base}\\ &= \frac{x^7y^5}{z^4} && \text{Laws 1 and 2}\end{aligned}$$

EXAMPLE

Simplifying Expressions with Negative Exponents

Eliminate negative exponents, and simplify each expression

(a) $\dfrac{6st^{-4}}{2s^{-2}t^2}$ **(b)** $\left(\dfrac{y}{3z^3}\right)^{-2}$

SOLUTION

(a) We use Law 7, which allows us to move a number raised to a power from the numerator to the denominator (or vice versa) by changing the sign of the exponent.

t^{-4} moves to denominator and becomes t^4

$$\frac{6st^{-4}}{2s^{-2}t^2} = \frac{6ss^2}{2t^2t^4} \qquad \text{Law 7}$$

s^{-2} moves to numerator and becomes s^2

$$= \frac{3s^3}{t^6} \qquad \text{Law 1}$$

(b) We use Law 6, which allows us to change the sign of the exponent of a fraction by inverting the fraction.

$$\left(\frac{y}{3z^3}\right)^{-2} = \left(\frac{3z^3}{y}\right)^2 \qquad \text{Law 6}$$

$$= \frac{9z^6}{y^2} \qquad \text{Laws 5 and 4}$$

EXAMPLE

3.1 Exponents: Evaluate expressions

a) $(-5)^3$

b) -5^3

c) $(-5)^2 \cdot \left(\frac{2}{5}\right)^2$

d) $-2^3 \cdot (-2)^0$

e) $-2^{-3} \cdot (-2)^0$

f) $\left(\frac{-3}{5}\right)^{-3}$

ACTIVITY 1.1

1.1 Exponents: Evaluate expressions

a) (-5^3) = (-5) × (-5) × (-5) = -125

b) -5^3 = - (5^3) = -(125) = -125

c) $(-5)^2 \times (\frac{2}{5})^2 = 25 \times \frac{4}{25} = 4$

d) $-2^3 \times (-2)^0 = -8$

e) $-2^{-3} \times (-2)^0 = -\frac{1}{8}$

f) $(\frac{-3}{5})^{-3} = -\frac{125}{27}$

SOLUTION

1.2 Exponents: Simplify expressions

Simplify each expression and eliminate any negative exponents.

a) $y^5 . y^2$

b) $(8x)^2$

c) $x^4 x^{-3}$

d) $y^{-5} . y^2$

e) $z^5\, z^{-3}\, z^{-4}$

f) $\frac{y^7 . y^0}{y^{10}}$

g) $\frac{z^2 . z^4}{z^3 z^{-1}}$

h) $(2a^3 a^2)^4$

i) $(-3z^2)^3(2z^3)$

ACTIVITY 1.2

1.2 Exponents: Simplify expressions

a) $y^5 y^2 = y^7$

b) $(8x)^2 = 8^2 x^2 = 64x^2$

c) $x^4 x^{-3} = x^{4-3} = x^1 = x$

d) $y^2 y^{-5} = y^{2-5} = y^{-3} = \frac{1}{y^3}$

e) $z^5 z^{-3} z^{-4} = z^{5-3-4} = z^{-2} = \frac{1}{z^2}$

f) $\frac{y^7 y^0}{y^{10}} = y^{7+0-10} = y^{-3} = \frac{1}{y^3}$

g) $\frac{z^2 z^4}{z^3 z^{-1}} = \frac{z^{2+4}}{z^{3-1}} = \frac{z^6}{z^2} = z^{6-2} = z^4$

h) $(2a^3 a^2)^4 = (2a^{3+2})^4 = (2a^5)^4 = 2^4 a^{5\times4} = 16a^{20}$

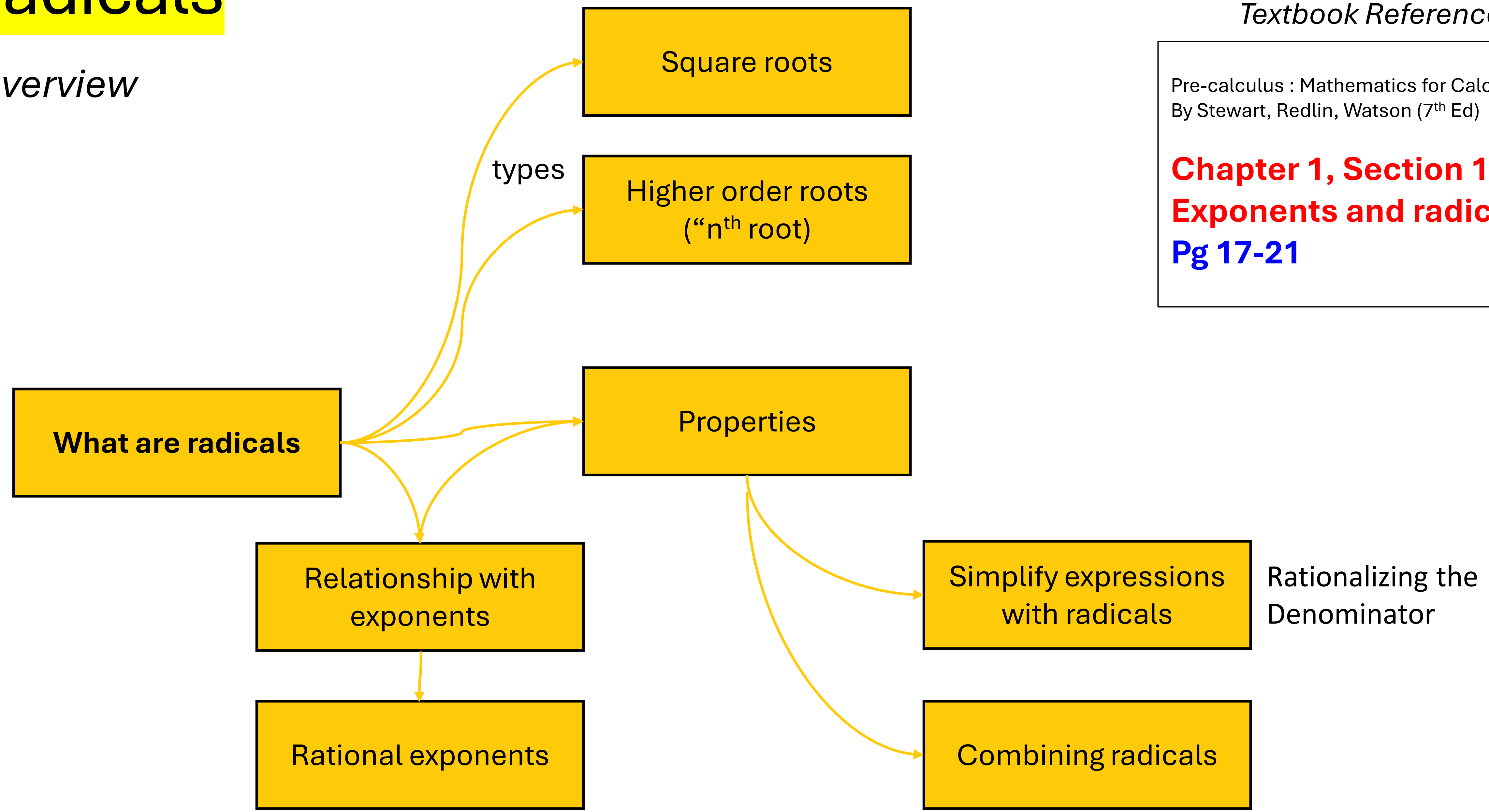
i) $(-3z^2)^3(2z^3) = -3^3 z^{2\times3}(2z^3) = -27\times2z^6 z^3 = -54z^{6+3} = -54z^9$

SOLUTION

2. Radicals

Radicals

Overview



Textbook Reference

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7^{th} Ed)

Chapter 1, Section 1.2: Exponents and radicals

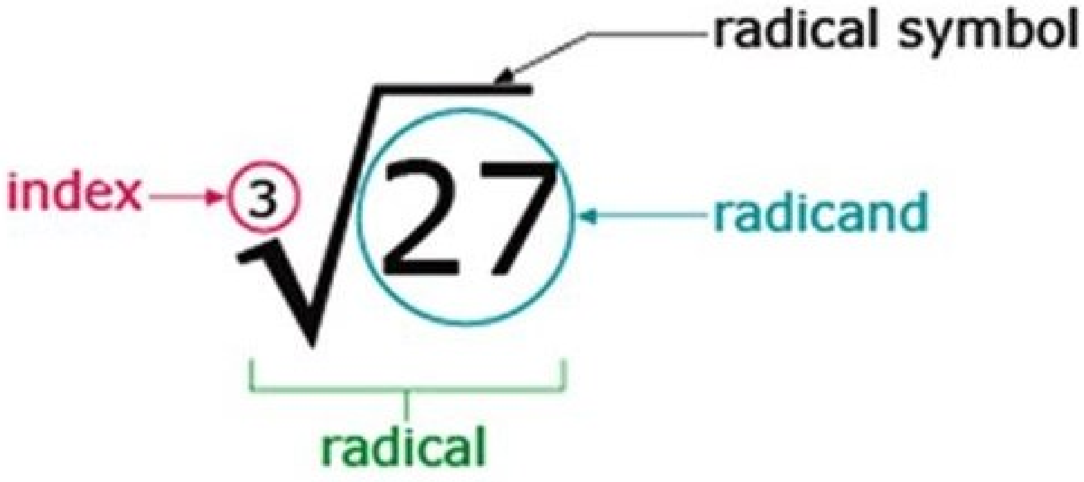
Pg 17-21

What is a **radical**?

$\sqrt{}$	means **"the positive square root of"** aka **"principal square root"**

$$\sqrt{x^2} = |x|$$

$$\sqrt{4} = 2$$



When we are asked to **determine a variable** such as the following example, we **include negative roots** too

$$x^2 - 4 = 0$$
$$x^2 = 4$$
$$\pm\sqrt{x^2} = \pm\sqrt{4}$$
$$x = \pm 2$$

What is *n*th Root?

DEFINITION OF *n*th ROOT

If n is any positive integer, then the **principal *n*th root** of a is defined as follows:

$$\sqrt[n]{a} = b \qquad \text{means} \qquad b^n = a$$

If n is even, we must have $a \geq 0$ and $b \geq 0$.

PROPERTIES OF *n*th ROOTS

Property	**Example**
1. $\sqrt[n]{ab} = \sqrt[n]{a}\sqrt[n]{b}$	$\sqrt[3]{-8 \cdot 27} = \sqrt[3]{-8}\sqrt[3]{27} = (-2)(3) = -6$
2. $\sqrt[n]{\dfrac{a}{b}} = \dfrac{\sqrt[n]{a}}{\sqrt[n]{b}}$	$\sqrt[4]{\dfrac{16}{81}} = \dfrac{\sqrt[4]{16}}{\sqrt[4]{81}} = \dfrac{2}{3}$
3. $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$	$\sqrt{\sqrt[3]{729}} = \sqrt[6]{729} = 3$
4. $\sqrt[n]{a^n} = a$ if n is odd	$\sqrt[3]{(-5)^3} = -5, \quad \sqrt[5]{2^5} = 2$
5. $\sqrt[n]{a^n} = \lvert a \rvert$ if n is even	$\sqrt[4]{(-3)^4} = \lvert -3 \rvert = 3$

Simplifying expressions involving nth Roots

(a) $\sqrt[3]{x^4} = \sqrt[3]{x^3x}$ Factor out the largest cube

$= \sqrt[3]{x^3}\sqrt[3]{x}$ Property 1: $\sqrt[3]{ab} = \sqrt[3]{a}\sqrt[3]{b}$

$= x\sqrt[3]{x}$ Property 4: $\sqrt[3]{a^3} = a$

(b) $\sqrt[4]{81x^8y^4} = \sqrt[4]{81}\sqrt[4]{x^8}\sqrt[4]{y^4}$ Property 1: $\sqrt[4]{abc} = \sqrt[4]{a}\sqrt[4]{b}\sqrt[4]{c}$

$= 3\sqrt[4]{(x^2)^4}|y|$ Property 5: $\sqrt[4]{a^4} = |a|$

$= 3x^2|y|$ Property 5: $\sqrt[4]{a^4} = |a|, |x^2| = x^2$

EXAMPLE

Combining Radicals

(a) $\sqrt{32} + \sqrt{200} = \sqrt{16 \cdot 2} + \sqrt{100 \cdot 2}$ — Factor out the largest squares

$= \sqrt{16}\sqrt{2} + \sqrt{100}\sqrt{2}$ — Property 1: $\sqrt{ab} = \sqrt{a}\sqrt{b}$

$= 4\sqrt{2} + 10\sqrt{2} = 14\sqrt{2}$ — Distributive Property

(b) If $b > 0$, then

$\sqrt{25b} - \sqrt{b^3} = \sqrt{25}\sqrt{b} - \sqrt{b^2}\sqrt{b}$ — Property 1: $\sqrt{ab} = \sqrt{a}\sqrt{b}$

$= 5\sqrt{b} - b\sqrt{b}$ — Property 5, $b > 0$

$= (5 - b)\sqrt{b}$ — Distributive Property

(c) $\sqrt{49x^2 + 49} = \sqrt{49(x^2 + 1)}$ — Factor out the perfect square

$= 7\sqrt{x^2 + 1}$ — Property 1: $\sqrt{ab} = \sqrt{a}\sqrt{b}$

EXAMPLE

Rational exponents

$$a^{1/n} = \sqrt[n]{a}$$

$$\sqrt{x} = x^{\frac{1}{2}}$$

$$\sqrt[3]{x} = x^{\frac{1}{3}}$$

$$\sqrt[4]{x} = x^{\frac{1}{4}}$$

DEFINITION OF RATIONAL EXPONENTS

For any rational exponent m/n in lowest terms, where m and n are integers and $n > 0$, we define

$$a^{m/n} = (\sqrt[n]{a})^m \qquad \text{or equivalently} \qquad a^{m/n} = \sqrt[n]{a^m}$$

If n is even, then we require that $a \geq 0$.

Using the Definition of Rational Exponents

(a) $4^{1/2} = \sqrt{4} = 2$

(b) $8^{2/3} = (\sqrt[3]{8})^2 = 2^2 = 4$ Alternative solution: $8^{2/3} = \sqrt[3]{8^2} = \sqrt[3]{64} = 4$

(c) $125^{-1/3} = \dfrac{1}{125^{1/3}} = \dfrac{1}{\sqrt[3]{125}} = \dfrac{1}{5}$

EXAMPLE

Using the Laws of Exponents with Rational Exponents

(a) $a^{1/3}a^{7/3} = a^{8/3}$ — Law 1: $a^m a^n = a^{m+n}$

(b) $\dfrac{a^{2/5}a^{7/5}}{a^{3/5}} = a^{2/5+7/5-3/5} = a^{6/5}$ — Law 1, Law 2: $\dfrac{a^m}{a^n} = a^{m-n}$

(c)

$$
\begin{aligned}
(2a^3b^4)^{3/2} &= 2^{3/2}(a^3)^{3/2}(b^4)^{3/2} && \text{Law 4: } (abc)^n = a^n b^n c^n \\
&= (\sqrt{2})^3 a^{3(3/2)} b^{4(3/2)} && \text{Law 3: } (a^m)^n = a^{mn} \\
&= 2\sqrt{2}a^{9/2}b^6
\end{aligned}
$$

(d)

$$
\begin{aligned}
\left(\frac{2x^{3/4}}{y^{1/3}}\right)^3\left(\frac{y^4}{x^{-1/2}}\right) &= \frac{2^3(x^{3/4})^3}{(y^{1/3})^3}\cdot(y^4x^{1/2}) && \text{Laws 5, 4, and 7} \\
&= \frac{8x^{9/4}}{y}\cdot y^4x^{1/2} && \text{Law 3} \\
&= 8x^{11/4}y^3 && \text{Laws 1 and 2}
\end{aligned}
$$

EXAMPLE

Simplifying by Writing Radicals as Rational Exponents

(a) $\dfrac{1}{\sqrt[3]{x^4}} = \dfrac{1}{x^{4/3}} = x^{-4/3}$ — Definition of rational and negative exponents

(b) $(2\sqrt{x})(3\sqrt[3]{x}) = (2x^{1/2})(3x^{1/3})$ — Definition of rational exponents

$= 6x^{1/2+1/3} = 6x^{5/6}$ — Law 1

(c) $\sqrt{x\sqrt{x}} = (xx^{1/2})^{1/2}$ — Definition of rational exponents

$= (x^{3/2})^{1/2}$ — Law 1

$= x^{3/4}$ — Law 3

EXAMPLE

Rationalizing Denominator

Put each fractional expression into standard form by rationalizing the denominator.

(a) $\dfrac{2}{\sqrt{3}}$ (b) $\dfrac{1}{\sqrt[3]{5}}$ (c) $\sqrt[7]{\dfrac{1}{a^2}}$

SOLUTION

This equals 1

(a) $\dfrac{2}{\sqrt{3}} = \dfrac{2}{\sqrt{3}} \cdot \dfrac{\sqrt{3}}{\sqrt{3}}$ Multiply by $\dfrac{\sqrt{3}}{\sqrt{3}}$

$= \dfrac{2\sqrt{3}}{3}$ $\sqrt{3} \cdot \sqrt{3} = 3$

(b) $\dfrac{1}{\sqrt[3]{5}} = \dfrac{1}{\sqrt[3]{5}} \cdot \dfrac{\sqrt[3]{5^2}}{\sqrt[3]{5^2}}$ Multiply by $\dfrac{\sqrt[3]{5^2}}{\sqrt[3]{5^2}}$

$= \dfrac{\sqrt[3]{25}}{5}$ $\sqrt[3]{5} \cdot \sqrt[3]{5^2} = \sqrt[3]{5^3} = 5$

(c) $\sqrt[7]{\dfrac{1}{a^2}} = \dfrac{1}{\sqrt[7]{a^2}}$ Property 2: $\sqrt[n]{\dfrac{a}{b}} = \dfrac{\sqrt[n]{a}}{\sqrt[n]{b}}$

$= \dfrac{1}{\sqrt[7]{a^2}} \cdot \dfrac{\sqrt[7]{a^5}}{\sqrt[7]{a^5}}$ Multiply by $\dfrac{\sqrt[7]{a^5}}{\sqrt[7]{a^5}}$

$= \dfrac{\sqrt[7]{a^5}}{a}$ $\sqrt[7]{a^2} \cdot \sqrt[7]{a^5} = a$

eliminate the radical in a denominator
by **multiplying both numerator and denominator**
by an appropriate expression

EXAMPLE

2.1 Radicals and Exponents

5.1 Write each radical expression using exponents, and each exponential expression using radicals.

	Radical expression	Exponential expression
a)		$10^{-3/2}$
b)		$2^{-1.5}$
c)	$\frac{1}{\sqrt{x^5}}$	

2.1 Radicals and Exponents

	Radical expression	Exponential expression
a)	$10^{-3/2} = \frac{1}{10^{3/2}} = \frac{1}{\sqrt[2]{(10^3)}} = \frac{1}{\sqrt{(10^3)}}$	$10^{-3/2}$
b)	$2^{-1.5} = \frac{1}{2^{1.5}} = \frac{1}{2^{\frac{3}{2}}} = \frac{1}{\sqrt{(2^3)}}$	$2^{-1.5}$
c)	$\frac{1}{\sqrt{x^5}}$	$\frac{1}{\sqrt{x^5}} = \frac{1}{x^{5\,1/2}} = \frac{1}{x^{(5)(1/2)}} = \frac{1}{x^{(5/2)}} = x^{-\frac{5}{2}}$

SOLUTION

2.2 Evaluate the following expressions

a) $2\sqrt[3]{81}$

b) $\dfrac{\sqrt{18}}{\sqrt{25}}$

c) $\sqrt{\frac{12}{49}}$

d) $\sqrt{10}\sqrt{32}$

e) $\dfrac{\sqrt{54}}{\sqrt{6}}$

ACTIVITY 2.2

2.2 Evaluate

a) $$2\sqrt[3]{81} = 2 \times (9^2)^{\frac{1}{3}} = 2 \times (3^4)^{\frac{1}{3}} = 6\sqrt[3]{3}$$

b) $$\frac{\sqrt{18}}{\sqrt{25}} = \frac{\sqrt{9} \times \sqrt{2}}{\sqrt{25}} = \frac{3\sqrt{2}}{5}$$

c) $$\sqrt{\frac{12}{49}} = \frac{\sqrt{4 \times 3}}{\sqrt{49}} = \frac{\sqrt{4} \times \sqrt{3}}{7} = \frac{2\sqrt{3}}{7}$$

d) $$\begin{aligned} &\sqrt{10}\sqrt{32} = \sqrt{10 \times 32} \\ &= \sqrt{320} \\ &= \sqrt{64} \times \sqrt{5} \\ &= 8\sqrt{5} \end{aligned}$$

e) $$\frac{\sqrt{54}}{\sqrt{6}} = \sqrt{9} = 3$$

SOLUTION

2.3 Simplify

Simplify the expression. Assume that the letters denote any positive real numbers

a) $\sqrt[5]{x^{10}}$

b) $\sqrt[3]{x^3y^6}$

c) $\sqrt{125} + \sqrt{45}$

d) $\sqrt[3]{54} - \sqrt[3]{16}$

e) $27^{1/3}$

f) $-\left(\frac{1}{8}\right)^{1/3}$

ACTIVITY 2.3

4.3 Simplify

a) $\sqrt[5]{x^{10}} = \sqrt[5]{(x^2)^5} = x^2$

b) $\sqrt[3]{x^3y^6} = \sqrt[3]{x^3} \times \sqrt[3]{\left(y^2\right)^3} = xy^2$

c)
$$\begin{aligned} &\sqrt{125} + \sqrt{45} = \sqrt{25}\sqrt{5} + \sqrt{9}\sqrt{5} \\ &= 5\sqrt{5} + 3\sqrt{5} \\ &= 8\sqrt{5} \end{aligned}$$

d)
$$\begin{aligned} &\sqrt[3]{54} - \sqrt[3]{16} = \sqrt[3]{27}\sqrt[3]{2} - \sqrt[3]{8}\sqrt[3]{2} \\ &= 3\sqrt[3]{2} - 2\sqrt[3]{2} = \sqrt[3]{2} \end{aligned}$$

e) $27^{1/3} = \sqrt[3]{27} = 3$

f) $-\left(\frac{1}{8}\right)^{1/3} = -\sqrt[3]{\frac{1}{8}} = -\frac{1}{2}$

SOLUTION

2.4 Standard form

Put each fractional expression into standard form by rationalizing the denominator

a) $\dfrac{12}{\sqrt{3}}$

b) $\sqrt{\dfrac{12}{5}}$

c) $\dfrac{8}{\sqrt[3]{5^2}}$

ACTIVITY 2.4

a) Multiply the numerator and the denominator by $\sqrt{3}$ and simplify:

$$\frac{12}{\sqrt{3}} = \frac{12}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{12\sqrt{3}}{\sqrt{3^2}} = \frac{12\sqrt{3}}{3} = 4\sqrt{3}$$

b) Rewrite as $\frac{\sqrt{12}}{\sqrt{5}}$:

$$\sqrt{\frac{12}{5}} = \frac{\sqrt{12}}{\sqrt{5}} = \ldots$$

Multiply the numerator and the denominator by $\sqrt{5}$ and simplify:

$$\ldots = \frac{\sqrt{12}}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{60}}{\sqrt{5^2}} = \frac{\sqrt{15 \cdot 4}}{5} = \frac{2\sqrt{15}}{5}$$

c) Multiply the numerator and the denominator by $\sqrt[3]{5}$ and simplify:

$$\frac{8}{\sqrt[3]{5^2}} = \frac{8}{\sqrt[3]{5^2}} \cdot \frac{\sqrt[3]{5}}{\sqrt[3]{5}} = \frac{8\sqrt[3]{5}}{\sqrt[3]{5^3}} = \frac{8\sqrt[3]{5}}{5}$$

SOLUTION